

1. ㉓	2. ㉔	3. ㉕	4. ㉖	5. ㉗
6. ㉘	7. ㉙	8. ㉚	9. ㉛	10. ㉜
11. ㉝	12. ㉞	13. ㉟	14. ㊱	15. ㊲
16. ㊳	17. ㊴	18. ㊵	19. ㊶	20. ㊷
21. ㊸	22. ㊹	23. ㊺	24. ㊻	25. ㊼
26. ㊽	27. ㊾	28. ㊿	29. $\frac{1}{2}(e-1)$	30. ㉑
31. ㉒	32. ㉓	33. ㉔	34. ㉕	35. ㉖
36. ㉗	37. ㉘	38. ㉙	39. ㉚	40. ㉛
41. ㉜	42. ㉝	43. ㉞	44. ㉟	45. ㊱
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1. $\frac{k}{n} = x, \frac{j}{m} = y$ 라 하면 $\frac{1}{n} = dx, \frac{1}{m} = dy$ 이므로

$$\begin{aligned} \lim_{m,n \rightarrow \infty} \sum_{k=1}^n \sum_{j=1}^m \frac{\pi k}{n} \sin\left(\frac{\pi j k}{4mn}\right) \frac{1}{mn} \\ = \int_0^1 \int_0^1 \pi x \sin\left(\frac{\pi}{4} xy\right) dy dx \\ = \int_0^1 -4 \left[\cos\left(\frac{\pi}{4} xy\right) \right]_0^1 dx \\ = -4 \int_0^1 \cos\left(\frac{\pi}{4} x\right) dx - 1 dx = 4\left(1 - \frac{2\sqrt{2}}{\pi}\right) \end{aligned}$$

2. $\iint_R \left(\frac{x}{2} + \frac{y}{4}\right) dx dy$

$$\begin{aligned} &= \int_0^{\pi} \int_{\frac{\pi}{2}}^{\pi} \frac{x}{2} e^{-\frac{y}{2}} dx dy \\ &= \int_0^{\pi} -2 \left[e^{-\frac{y}{2}} \right]_{\frac{\pi}{2}}^{\pi} e^{-\frac{y}{2}} dy \\ &= 2 \int_0^{\pi} e^{-\frac{y}{2}} dy = 4 \end{aligned}$$

3. $\iint_D x \sin y dx dy = \int_0^{\sqrt{\pi}} \int_0^{x^2} x \sin y dy dx$

$$\begin{aligned} &= \int_0^{\sqrt{\pi}} x [-\cos y]_0^{x^2} dx \\ &= \int_0^{\sqrt{\pi}} x (1 - \cos(x^2)) dx \\ &= \frac{1}{2} [x^2 - \sin(x^2)]_0^{\sqrt{\pi}} \\ &= \frac{\pi}{2} \end{aligned}$$

4. $\iint_R y^2 dx dy = \int_1^4 \int_{\frac{1}{3}x - \frac{1}{3}}^{\frac{1}{3}x + \frac{7}{3}} y^2 dy dx$

$$\begin{aligned} &= \int_1^4 \left[\frac{1}{3} y^3 \right]_{\frac{1}{3}x - \frac{1}{3}}^{\frac{1}{3}x + \frac{7}{3}} dx \\ &= \frac{1}{3} \int_1^4 \left(-\frac{1}{3}x + \frac{7}{3} \right)^3 dx \\ &= \frac{1}{3} \left[-\frac{3}{4} \left(-\frac{1}{3}x + \frac{7}{3} \right)^4 + \frac{3}{4} \left(\frac{1}{3}x - \frac{1}{3} \right)^4 \right]_1^4 \\ &= \frac{1}{3} \left[-\frac{3}{4} \left(\frac{3}{4} - 12 \right) \right] \\ &= \frac{7}{2} \end{aligned}$$

5. $\iint_R \frac{x^4}{y} \exp\left(\frac{x^2}{y}\right) dx dy$

$$\begin{aligned} &= \int_1^2 \int_{x^2}^{2x^2} \frac{x^4}{y} \exp\left(\frac{x^2}{y}\right) dy dx \quad \left(\frac{x^2}{y} = t \text{로 치환}\right) \\ &= \int_1^2 \int_{\frac{1}{2}}^1 t e^t dt dx \\ &= \int_1^2 t e^t dt \int_1^2 dx \\ &= [e^t t - e^t]_1^2 \times 1 \\ &= \frac{1}{2} e^{\frac{1}{2}} \end{aligned}$$

6. $\int_0^1 \int_{1-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\sqrt{2-y}} dy dx$

$$\begin{aligned} &= \int_0^2 \int_0^{\sqrt{2y-y^2}} \frac{1}{\sqrt{2-y}} dx dy \\ &= \int_0^2 \sqrt{2y-y^2} \frac{1}{\sqrt{2-y}} dy \\ &= \int_0^2 \sqrt{y} dy \\ &= \frac{2}{3} \left[y^{\frac{3}{2}} \right]_0^2 = \frac{4\sqrt{2}}{3} \end{aligned}$$

7. $\lim_{t \rightarrow 1} \int_{-1}^t \int_0^t \frac{1}{1-xy} dx dy$

$$\begin{aligned} &= \int_0^1 \int_0^1 \frac{1}{1-xy} dx dy \\ &= \int_0^1 -\frac{1}{y} [\ln(1-xy)]_0^1 dy \\ &= \int_0^1 -\frac{1}{y} \ln(1-y) dy \\ &= \int_0^1 \left(y + \frac{1}{2}y^2 + \frac{1}{3}y^3 + \dots \right) dy \\ &= \int_0^1 \left(1 + \frac{1}{2}y + \frac{1}{3}y^2 + \dots \right) dy \\ &= \left[y + \frac{1}{2^2}y^2 + \frac{1}{3^2}y^3 + \dots \right]_0^1 \\ &= 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \sum_{n=1}^{\infty} \frac{1}{n^2} \quad \therefore 1 \leq \sum_{n=1}^{\infty} \frac{1}{n^2} \leq 2 \end{aligned}$$

참고. $p > 1$ 일 때, $\frac{1}{p-1} \leq \sum_{n=1}^{\infty} \frac{1}{n^p} \leq \frac{p}{p-1}$ 이다.

8.

세 개의 점 $(0,0,1), \left(\frac{\pi}{2}, 0, 2\right), \left(\frac{\pi}{2}, \frac{\pi}{2}, 2\right)$ 을

$z = f(x,y) = a \sin x + b \cos y$ 에 대입하면

$1 = b, 2 = a + b, 2 = a$ 의 관계식을 얻는다.

연립방정식 $1 = b, 2 = a + b, 2 = a$ 을 행렬로 표현하면

$$Ax = b \Leftrightarrow \begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \quad \text{이므로 최소제곱해를 } x \text{라 할 때,}$$

$$x = (A^T A)^{-1} A^T b$$

$$= \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 4 \\ 3 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 5 \\ 5 \end{pmatrix}$$

이다. 그러므로 $f(x,y) = \frac{5}{3} \sin x + \frac{2}{3} \cos y$ 이고

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} f(x,y) dx dy &= \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \frac{5}{3} \sin x + \frac{2}{3} \cos y dx dy \\ &= \int_0^{\frac{\pi}{2}} \left[-\frac{5}{3} \cos x + \frac{2}{3} y \right]_0^{\frac{\pi}{2}} dy \\ &= \int_0^{\frac{\pi}{2}} \left(\frac{5}{3} + \frac{2}{3} y \right) dy = \frac{5}{6} \pi + \frac{\pi}{3} = \frac{7}{6} \pi \end{aligned}$$

9. 적분 순서의 변경에 의해 다음과 같다.

$$\begin{aligned} & \int_0^1 \int_{\frac{1}{2}\sin^{-1}y}^{\frac{\pi}{2}} \frac{1}{\cos^2 x + 1} dx dy \\ &= \int_0^{\frac{\pi}{2}} \int_0^{\sin 2x} \frac{1}{\cos^2 x + 1} dy dx \\ &= \int_0^{\frac{\pi}{2}} \frac{2 \sin x \cos x}{\cos^2 x + 1} dx \\ &= \int_{\frac{\sqrt{2}}{2}}^1 \frac{2t}{t^2 + 1} dt \quad (\cos x = t \text{ 치환}) \\ &= [\ln(t^2 + 1)]_{\frac{\sqrt{2}}{2}}^1 = \ln \frac{4}{3} \end{aligned}$$

10. 적분 순서의 변경에 의해 다음과 같다.

$$\begin{aligned} & \int_0^1 \int_{\sqrt{x}}^1 \sqrt{1+y^4} dy dx \\ &= \int_0^1 \int_0^{y^2} \sqrt{1+y^4} dx dy \\ &= \int_0^1 y^2 \sqrt{1+y^4} dy \\ &= \int_0^1 y^2 (1+y^4)^{\frac{1}{2}} dy \\ &= \frac{1}{4} \frac{2}{3} \left[(1+y^4)^{\frac{3}{2}} \right]_0^1 \\ &= \frac{1}{6} (2\sqrt{2} - 1) \end{aligned}$$

11. 적분 순서의 변경에 의해 다음과 같다.

$$\begin{aligned} & \int_0^{\frac{\pi}{2}} \int_{\frac{\sqrt{x}}{2}}^{\frac{\sqrt{x}}{2}} \frac{y \cos(x^2)}{x^3} dx dy \\ &= \int_0^{\frac{\sqrt{x}}{2}} \int_0^{x^2} \frac{y \cos(x^2)}{x^3} dy dx \\ &= \int_0^{\frac{\sqrt{x}}{2}} \frac{1}{2} x^4 \cdot \frac{\cos(x^2)}{x^3} dx \\ &= \frac{1}{2} \int_0^{\frac{\sqrt{x}}{2}} x \cos(x^2) dx \\ &= \frac{1}{4} [\sin(x^2)]_0^{\frac{\sqrt{x}}{2}} = \frac{1}{4} \sin\left(\frac{\pi}{4}\right) = \frac{1}{4\sqrt{2}} \end{aligned}$$

12. 미적분학의 기본정리에 의해

$$\begin{aligned} & \int_0^1 f'(x) dx = f(1) - f(0) \\ & \Leftrightarrow \int_0^1 \int_{\tan^{-1}x}^{\frac{\pi}{4}} 3\cos^2 t \sqrt{1+\sin^2 t} dt dx = f(1) - f(0) \text{ 이 성립한다.} \\ & \text{또한, } \int_0^1 \int_{\tan^{-1}x}^{\frac{\pi}{4}} 3\cos^2 t \sqrt{1+\sin^2 t} dt dx \\ &= \int_0^{\frac{\pi}{4}} \int_0^{\tan t} 3\cos^2 t \sqrt{1+\sin^2 t} dx dt \quad (\because \text{적분 순서 변경}) \\ &= \int_0^{\frac{\pi}{4}} 3 \tan t \cos^2 t \sqrt{1+\sin^2 t} dt dx \\ &= \int_0^{\frac{\pi}{4}} 3 \sin t \cos t \sqrt{1+\sin^2 t} dt \\ &= \frac{3}{2} \int_0^{\frac{\pi}{4}} 2 \sin t \cos t (1+\sin^2 t)^{\frac{1}{2}} dt \\ &= \frac{3}{2} \left[\frac{2}{3} (1+\sin^2 t)^{\frac{3}{2}} \right]_0^{\frac{\pi}{4}} \\ &= \left(\frac{3}{2} \right)^{\frac{3}{2}} - 1 = \frac{3\sqrt{6}}{4} - 1 \text{ 이므로} \\ & f(1) = \frac{3\sqrt{6}}{4} - 1 \text{ 이다. } (\because f(0) = 0) \end{aligned}$$

$$\begin{aligned} 13. \int_{-2}^0 \int_{-y}^2 e^{y^2} dy dx &= \int_{-2}^0 \int_{-y}^0 e^{y^2} dx dy \quad (\because \text{적분 순서 변경}) \\ &= \int_{-2}^0 y e^{y^2} dy = \frac{1}{2} e^{y^2} \Big|_{-2}^0 = \frac{1}{2} (e^4 - 1) \end{aligned}$$

$$14. \int_0^{\infty} \int_0^y y^{-\frac{3}{2}} e^{-x} dx dy = \int_0^{\infty} \int_y^{\infty} y^{-\frac{3}{2}} e^{-x} dy dx \text{ 성립한다.}$$

$$\begin{aligned} & \text{따라서 } \int_0^{\infty} \int_0^y y^{-\frac{3}{2}} e^{-x} dx dy \\ &= \int_0^{\infty} \int_x^{\infty} y^{-\frac{3}{2}} e^{-x} dy dx \\ &= \int_0^{\infty} \left[-2y^{-\frac{1}{2}} \right]_x^{\infty} e^{-x} dx \\ &= \int_0^{\infty} -2(0 - x^{-\frac{1}{2}}) e^{-x} dx \\ &= 2 \int_x^{\infty} x^{-\frac{1}{2}} e^{-x} dx \\ &= 2 \times \Gamma\left(\frac{1}{2}\right) = 2\sqrt{\pi} \end{aligned}$$

15. 적분 순서의 변경에 의해 다음과 같다.

$$\begin{aligned} & \int_0^1 \int_{\cos^{-1}y}^{\cos^{-1}y} \sin x \sqrt{1+\sin^2 x} dx dy \\ &= \int_0^{\frac{\pi}{2}} \int_0^{\cos x} \sin x \sqrt{1+\sin^2 x} dy dx \\ &= \int_0^{\frac{\pi}{2}} \sin x \cos x \sqrt{1+\sin^2 x} dx \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} 2 \sin x \cos x (1+\sin^2 x)^{\frac{1}{2}} dx \\ &= \frac{1}{2} \frac{2}{3} \left[(1+\sin^2 x)^{\frac{3}{2}} \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{3} (2\sqrt{2} - 1) \end{aligned}$$

$$\begin{aligned} 16. & \int_0^0 \int_{-1}^0 \sin\{(y-1)^2\} dy dx + \int_0^1 \int_{-1}^{-x} \sin\{(y-1)^2\} dy dx \\ &= \int_0^1 \int_{-1}^{-y} \sin\{(y-1)^2\} dx dy \\ &= \int_0^1 2(1-y) \sin\{(y-1)^2\} dx dy \\ &= - \int_0^1 2(y-1) \sin\{(y-1)^2\} dx dy \\ &= [\cos\{(y-1)^2\}]_0^1 \\ &= 1 - \cos 1 \end{aligned}$$

$$\begin{aligned} 17. & \int_{-1}^0 \int_{-\sqrt{y+1}}^{\sqrt{y+1}} e^{-2x^2+3x^2+12x} dx dy \\ &+ \int_0^3 \int_{-1}^{\sqrt{y+1}} e^{-2x^2+3x^2+12x} dx dy \\ &= \int_{-1}^2 \int_{x^2-1}^{x^2+1} e^{-2x^2+3x^2+12x} dy dx \\ &= \int_{-1}^2 (-x^2+x+2) e^{-2x^2+3x^2+12x} dx \\ &= \frac{1}{6} [e^{-2x^2+3x^2+12x}]_{-1}^2 \\ &= \frac{1}{6} (e^{20} - e^{-7}) \end{aligned}$$

18.

$$\begin{aligned}
& \int_0^1 \int_0^2 e^{y-x} dx dy \\
&= \int_0^1 \int_0^y e^{y-x} dx dy + \int_0^1 \int_y^2 e^{y-x} dx dy \\
&= \int_0^1 [-e^{-x}]_0^y dy + \int_0^1 [-e^{-x}]_y^2 dy \\
&= \int_0^1 (1-e^{-y})e^y dy + \int_0^1 (1-e^{-x})e^x dx \\
&= 2 \int_0^1 e^x - 1 dx \\
&= 2[e^x - x]_0^1 \\
&= 2(e^2 - 3)
\end{aligned}$$

19.

$$\begin{aligned}
& \int_0^1 \int_0^1 e^{x+y} dy dx \\
&= \int_0^1 \int_0^{1-x} e^{x+y} dy dx + \int_0^1 \int_{1-x}^1 e^{x+y} dy dx \\
&= \iint_{D_1} 1 dy dx + e \iint_{D_2} 1 dy dx \\
&= D_1 \text{의 넓이} + e \times D_2 \text{의 넓이} \\
&= \frac{1}{2} + \frac{e}{2}
\end{aligned}$$

20.

$$\begin{aligned}
D_1 : x^2 + y^2 < 1, x \geq 0, y \geq 0 \\
D_2 : 1 \leq x^2 + y^2 < 2, 0 \leq x \leq 1.0 \leq y \leq 1 \\
\int_0^1 \int_0^1 [x^2 + y^2] dx dy \\
&= \iint_{D_1} 0 dx dy + \iint_{D_2} 1 dx dy \\
&= D_2 \text{의 넓이} = 1 - \frac{\pi}{4}
\end{aligned}$$

21.

$$\begin{aligned}
D_0 : 0 \leq x^2 + y^2 < 1, \\
D_1 : 1 \leq x^2 + y^2 < 2 \\
\vdots \\
D_9 : 9 \leq x^2 + y^2 < 10
\end{aligned}$$

$$\begin{aligned}
& \iint_D [x^2 + y^2] dA \\
&= \iint_{D_0} 0 dA + \iint_{D_1} 1 dA + \iint_{D_2} 2 dA + \cdots + \iint_{D_9} 9 dA \\
&= \pi + 2\pi + \cdots + 9\pi \\
&= \frac{9 \times 10}{2} \pi = 45\pi
\end{aligned}$$

참고. $D : x^2 + y^2 \leq n$ (n 은 자연수) 일 때,

$$\iint_D [x^2 + y^2] dA = \frac{n(n-1)}{2} \pi \text{이다.}$$

22.

$$\begin{aligned}
& \int_0^1 \int_0^1 e^{\max(y^2, 4x^2)} dy dx \\
&= \int_0^1 \int_0^{\frac{y}{2}} e^{y^2} dx dy + \int_0^1 \int_{\frac{y}{2}}^1 e^{4x^2} dy dx \\
&= \int_0^1 \frac{y}{2} e^{y^2} dy + \int_0^1 2xe^{4x^2} dx \\
&= \frac{1}{4} [e^{y^2}]_0^1 + \frac{1}{4} [e^{4x^2}]_0^1 \\
&= \frac{1}{2} (e^{16} - 1)
\end{aligned}$$

23.

$$\begin{aligned}
f(x) &= \int_0^{x^2} \int_{\sqrt{x}}^1 e^{t^2+s} dt ds \\
&= \int_0^{x^2} \int_{\sqrt{x}}^1 e^t e^s dt ds \\
&= \int_0^{x^2} e^s ds \int_{\sqrt{x}}^1 e^t dt \\
f'(x) &= e^{x^2} 2x \int_{\sqrt{x}}^1 e^t dt - \int_0^{x^2} e^t ds e^x \cdot \frac{1}{2\sqrt{x}} \\
f'(1) &= -\frac{e}{2} \int_0^1 e^t ds = -\frac{e}{2} (e-1)
\end{aligned}$$

24.

$$\begin{aligned}
F(t) &= \int_0^t \int_2^4 2 \cos y^2 dy dx \\
&= \int_0^t \int_0^y 2 \cos y^2 dx dy \\
&= \int_0^t 2y \cos y^2 dy \\
F'(t) &= 2t \cos t^2, \quad F'\left(\frac{\sqrt{\pi}}{2}\right) = \frac{\sqrt{\pi}}{\sqrt{2}} \text{이다.}
\end{aligned}$$

25.

연쇄법칙에 의해

$$\begin{aligned}
\frac{\partial (g_1, g_2, g_3)}{\partial (x, y, z)} &= \frac{\partial (g_1, g_2, g_3)}{\partial (u, v)} \cdot \frac{\partial (u, v)}{\partial (x, y, z)} \\
&= \begin{pmatrix} \frac{\partial g_1}{\partial u} & \frac{\partial g_1}{\partial v} \\ \frac{\partial g_2}{\partial u} & \frac{\partial g_2}{\partial v} \\ \frac{\partial g_3}{\partial u} & \frac{\partial g_3}{\partial v} \end{pmatrix} \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \\ \frac{\partial f_3}{\partial x} & \frac{\partial f_3}{\partial y} & \frac{\partial f_3}{\partial z} \end{pmatrix} \text{이 성립한다.} \\
(x, y, z) &= (1, 1, 1) \text{ 일 때, } (u, v) = (2, 3) \text{ 이고} \\
\begin{pmatrix} \frac{\partial g_1}{\partial u} & \frac{\partial g_1}{\partial v} \\ \frac{\partial g_2}{\partial u} & \frac{\partial g_2}{\partial v} \\ \frac{\partial g_3}{\partial u} & \frac{\partial g_3}{\partial v} \end{pmatrix} &= \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \\ \frac{\partial f_3}{\partial x} & \frac{\partial f_3}{\partial y} & \frac{\partial f_3}{\partial z} \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix} \text{이므로} \\
\frac{\partial (g_2, v, F)}{\partial x, y} \bigg|_{(1,1,1)} &= 6 + 2 = 8 \text{이다.}
\end{aligned}$$

26.

 $x+y=u, y-x=v$ 으로 변수변환을 이용하면

$$\begin{aligned}
|J| &= \frac{1}{\begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix}} = \frac{1}{2} \text{이고} \\
& \iint_R \sqrt{y-x} (x+y)^2 dx dy \\
&= \int_0^1 \int_0^1 \sqrt{v} u^2 \frac{1}{2} du dv = \frac{1}{2} \int_0^1 \sqrt{v} \frac{1}{3} v^2 dv \\
&= \frac{1}{6} \int_0^1 \sqrt{v} v^2 dv = \frac{1}{6} \times \frac{2}{3} = \frac{1}{9} \text{이다.}
\end{aligned}$$

27.

 $u=2x+y, v=x-y$ 으로 변수변환을 이용하면

$$\begin{aligned}
|J| &= \frac{1}{\begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix}} = \frac{1}{3} \text{이고} \\
& \iint_R (2x^2 - xy - y^2) dx dy \\
&= \iint_R (2x+y)(x-y) dx dy \\
&= \frac{1}{3} \int_{-1}^2 \int_{-1}^1 uv du dv = \frac{1}{6} \int_{-1}^2 [u^2]_{-1}^1 v dv \\
&= \frac{33}{12} [v^2]_{-1}^2 = \frac{33}{4} \text{이다.}
\end{aligned}$$

28.

 $x+y=u, x-y=v$ 으로 변수변환을 이용하면

$$|J| = \left| \frac{1}{1-1} \right| = \frac{1}{2} \text{ 이고}$$

$$\begin{aligned} & \iint_D \cosh\left(\frac{x-y}{x+y}\right) dx dy \\ &= \frac{1}{2} \int_1^3 \int_{-u}^u \cosh\left(\frac{v}{u}\right) dv du \\ &= \int_1^3 \int_0^u \cosh\left(\frac{v}{u}\right) dv du = \int_1^3 u \left\{ \sinh\left(\frac{v}{u}\right) \right\}_0^u du \\ &= \int_1^3 u (\sinh 1 - 0) du = \sinh 1 \int_1^3 u du \\ &= 4 \sinh 1 = 4 \cdot \frac{e^1 - e^{-1}}{2} = 2(e - e^{-1}) \text{ 이다.} \end{aligned}$$

29.

 $x-y=u, x-2y=v$ 으로 변수변환을 이용하면

$$|J| = \left| \frac{1}{1-2} \right| = 1 \text{ 이고}$$

$$\begin{aligned} & \iint_A (x-y)^2 dx dy \\ &= \int_0^1 \int_0^u e^{v^2} dv du \\ &= \int_0^1 u e^{u^2} du \\ &= \frac{1}{2} [e^{u^2}]_0^1 \\ &= \frac{1}{2} (e-1) \text{ 이다.} \end{aligned}$$

30.

 $x-3y=u, 2x+3y=v$ 으로 변수변환을 이용하면

$$\begin{aligned} |J| &= \left| \frac{1}{2-3} \right| = \frac{1}{9} \\ & \iint_D (x-3y-1)e^{2x+3y} \cos(x-3y) dA \\ &= \frac{1}{9} \int_0^3 \int_0^1 (u-1)e^v \cos u dv du \\ &= \frac{1}{9} \int_0^3 e^v dv \int_0^1 (u-1) \cos u du \\ &= \frac{1}{9} [e^v]_0^3 \times [\sin u(u-1) + \cos u]_0^1 \\ &= \frac{1}{9} (e^3 - 1)(\cos 1 - 1) \end{aligned}$$

31.

두 곡선 $x^2=2y, x^2=3y$ 과 두 직선 $y=4x, y=5x$ 로 둘러싸인 영역을 D 라 할 때, 영역 D 의 넓이는 $\iint_D 1 dA$ 이다.

$$\begin{aligned} \frac{x^2}{y} &= u, \frac{y}{x} = v \text{으로 변수변환을 이용하면} \\ J &= \frac{1}{\begin{vmatrix} \frac{2x}{y} & -\frac{x^2}{y^2} \\ -\frac{y}{x^2} & \frac{1}{x} \end{vmatrix}} = \frac{1}{\frac{2}{y} - \frac{1}{y}} = y = uv^2 \text{ 이다.} \end{aligned}$$

$$\text{따라서 } \iint_D 1 dA = \int_2^3 \int_1^5 uv^2 dv du = \frac{305}{6} \text{ 이다.}$$

32. $xy=u, \frac{y}{x}=v$ 으로 변수변환을 이용하면

$$J = \frac{1}{\begin{vmatrix} \frac{y}{x} & x \\ -\frac{y}{x^2} & \frac{1}{x} \end{vmatrix}} = \frac{1}{\frac{y}{x} + \frac{y}{x}} = \frac{1}{2\frac{y}{x}} = \frac{1}{2v} \text{ 이다.}$$

$$\begin{aligned} \text{따라서 } & \iint_R \left(\sqrt{\frac{y}{x}} + \sqrt{xy} \right) dx dy \\ &= \int_1^4 \int_1^9 (\sqrt{v} + \sqrt{u}) \frac{1}{2v} du dv \end{aligned}$$

$$\begin{aligned} &= \frac{1}{2} \int_1^4 \int_1^9 \left(\frac{1}{\sqrt{v}} + \frac{\sqrt{u}}{v} \right) du dv \\ &= \frac{1}{2} \int_1^4 \left[\frac{u}{\sqrt{v}} + \frac{2}{3} \frac{u^{\frac{3}{2}}}{v} \right]_1^9 dv \\ &= \frac{1}{2} \int_1^4 \left(\frac{8}{\sqrt{v}} + \frac{2}{3} \frac{26}{v} \right) dv \\ &= \frac{1}{2} \left[16\sqrt{v} + \frac{52}{3} \ln v \right]_1^4 \\ &= \frac{1}{2} \left(16 + \frac{52}{3} \ln 4 \right) \\ &= 8 + \frac{52}{3} \ln 2 \end{aligned}$$

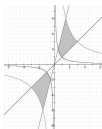
33. $x^2-y^2=u, xy=v$ 으로 변수변환 이용하면

$$J = \frac{1}{\begin{vmatrix} \frac{2x}{y} & -2y \\ y & 2(x^2+y^2) \end{vmatrix}} = \frac{1}{2(x^2+y^2)} \text{ 이고}$$

$$\begin{aligned} & \iint_D \frac{x^4-y^4}{1+xy} dx dy \\ &= \iint_D \frac{(x^2-y^2)(x^2+y^2)}{1+xy} dx dy \\ &= \iint_D \frac{u(x^2+y^2)}{1+v} \cdot \frac{1}{2(x^2+y^2)} du dv \\ &= \frac{1}{2} \iint_D \frac{u}{1+v} du dv \\ &= \frac{1}{2} \int_0^1 \int_0^1 \frac{u}{1+v} du dv \\ &= \frac{1}{2} \int_0^1 \left[\frac{1}{2} u^2 \right]_0^1 dv \\ &= \frac{1}{4} \int_0^1 \frac{1}{1+v} dv \\ &= \frac{1}{4} [\ln(1+v)]_0^1 \\ &= \frac{1}{4} \ln 2 \end{aligned}$$

34. $xy=u, \frac{y}{x}=v$ 으로 변수변환을 이용하면

$$J = \frac{1}{\begin{vmatrix} \frac{y}{x} & x \\ -\frac{y}{x^2} & \frac{1}{x} \end{vmatrix}} = \frac{1}{2\frac{y}{x}} = \frac{1}{2v} \text{ 이고}$$

영역 R 은 변수변환 $xy=u, \frac{y}{x}=v$ 에 대하여 일대일을 만족하지 못한다.

이때, $f(x,y) = \left(\frac{y}{x}\right)^2 e^{\left(\frac{y}{x}\right)^2}$ 은 1사분면에 있는 영역 R_1 과 3사분면에 있는 영역 R_2 에 대하여 높이가 같기 때문에 적분 값은 R_1 에서 계산한 값에 두배를 하면 되고 R_1 으로부터 영역을 제한하면 변수변환에 대하여 일대일을 만족한다. 따라서

$$\begin{aligned} & \iint_R \left(\frac{y}{x}\right)^2 e^{\left(\frac{y}{x}\right)^2} dA \\ &= 2 \iint_{R_1} \left(\frac{y}{x}\right)^2 e^{\left(\frac{y}{x}\right)^2} dA \\ &= 2 \int_1^4 \int_1^9 v^2 e^{v^2} \frac{1}{2v} du dv \\ &= \int_1^4 \int_1^9 v e^{v^2} du dv = \int_1^4 8 v e^{v^2} dv = [4e^{v^2}]_1^4 = 4(e^{16} - 1) \end{aligned}$$

35.

$$\begin{aligned}
 & \int_1^2 \int_0^{\sqrt{2x-x^2}} \frac{1}{(x^2+y^2)^2} dy dx \\
 &= \int_0^{\frac{\pi}{4}} \int_{\sec\theta}^{\frac{2}{\cos\theta}} \frac{r}{r^4} dr d\theta = -\frac{1}{2} \int_0^{\frac{\pi}{4}} [r^{-2}]_{\sec\theta}^{\frac{2}{\cos\theta}} d\theta \\
 &= -\frac{1}{2} \int_0^{\frac{\pi}{4}} \left(\frac{1}{4} \sec^2\theta - \cos^2\theta \right) d\theta \\
 &= -\frac{1}{8} [\tan\theta]_0^{\frac{\pi}{4}} + \frac{1}{4} \int_0^{\frac{\pi}{4}} (1 + \cos 2\theta) d\theta \\
 &= -\frac{1}{8} + \frac{1}{4} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{4}} \\
 &= \frac{\pi}{16}
 \end{aligned}$$

36.

$$\begin{aligned}
 & \int_0^2 \int_{-\sqrt{2x-x^2}}^0 (x^2+y^2)^{\frac{3}{2}} dy dx \\
 &= \int_{-\frac{\pi}{2}}^0 \int_0^{2\cos\theta} (r^2)^{\frac{3}{2}} \cdot r dr d\theta \\
 &= \int_{-\frac{\pi}{2}}^0 \int_0^{2\cos\theta} r^4 dr d\theta = \int_{-\frac{\pi}{2}}^0 \frac{1}{5} [r^5]_0^{2\cos\theta} d\theta \\
 &= \frac{32}{5} \int_{-\frac{\pi}{2}}^0 \cos^5\theta d\theta = \frac{32}{5} \int_0^{\frac{\pi}{2}} \cos^5\theta d\theta \quad (\because \text{우함수}) \\
 &= \frac{32}{5} \times \left(\frac{4}{5} \cdot \frac{2}{3} \right) \quad (\because \text{wallis 공식}) \\
 &= \frac{256}{75}
 \end{aligned}$$

37.

$$\begin{aligned}
 & \iint_D \frac{\ln(x^2+y^2)}{(x^2+y^2)^2} dy dx \\
 &= \int_0^{2\pi} \int_1^{\frac{1}{\cos\theta}} \frac{\ln(r^2)}{r^4} r dr d\theta \\
 &= \int_0^{2\pi} \int_1^{\frac{1}{\cos\theta}} \frac{2\ln(r)}{r^3} dr d\theta \\
 &= \int_0^{2\pi} \int_1^{2\pi-3\ln(r)} r dr d\theta \\
 &= 2\pi \times 2 \left[-\frac{1}{2} r^{-2} \ln r - \frac{1}{4} r^{-2} \right]_1^{\frac{1}{\cos\theta}} \\
 &= \pi \left(1 - \frac{3}{e} \right)
 \end{aligned}$$

38.

$$\begin{aligned}
 & \iint_D \frac{1}{\sqrt{x^2+y^2}} dx dy \\
 &= \int_0^{\frac{\pi}{4}} \int_{2\sec\theta}^{\frac{1}{\cos\theta}} \frac{1}{r} r dr d\theta \\
 &= \int_0^{\frac{\pi}{4}} 2\sec\theta - 2\cos\theta d\theta \\
 &= [2\ln(\sec\theta + \tan\theta) - 2\sin\theta]_0^{\frac{\pi}{4}} \\
 &= 2\ln(\sqrt{2}+1) - \sqrt{2}
 \end{aligned}$$

39.

$$\begin{aligned}
 & \iint_D \ln(x^2+y^2) dx dy \\
 &= \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \int_0^1 r \ln(r^2) r dr d\theta \\
 &= 2 \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \int_0^1 r \ln(r) r dr d\theta \\
 &= 2 \int_0^1 r \ln(r) dr \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} d\theta \\
 &= 2 \left[\frac{1}{2} r^2 \ln r - \frac{1}{4} r^2 \right]_0^1 \left(\frac{\pi}{2} - \beta \right) = \frac{\beta}{2} - \frac{\pi}{4}
 \end{aligned}$$

40.

$$\begin{aligned}
 & \int_{-\frac{1}{\sqrt{2}}}^0 \left(\int_{-x}^{\sqrt{1-x^2}} y dy \right) dx + \int_0^{\frac{1}{2}} \left(\int_0^{\sqrt{3}y} y dx \right) dy + \int_{\frac{1}{2}}^1 \left(\int_0^{\sqrt{1-y^2}} y dx \right) dy \\
 &= \int_{\frac{\pi}{6}}^{\frac{3\pi}{4}} \int_0^1 r^2 \sin\theta dr d\theta \\
 &= \frac{1}{3} \left(\frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \right)
 \end{aligned}$$

41.

$$\begin{aligned}
 & D: \{(x, y) | (x^2+y^2)^2 - (x^2-y^2) \leq 0\} \\
 & \Leftrightarrow D: \{(r, \theta) | r^4 - r^2 (\cos^2\theta - \sin^2\theta) \leq 0\} \\
 & \Leftrightarrow D: \{(r, \theta) | r^2 \leq \cos 2\theta\} \text{ 이므로} \\
 & \iint_D \frac{1}{(1+x^2+y^2)^2} dy dx \\
 &= \iint_D \frac{r}{(1+r^2)^2} dr d\theta \\
 &= \int_0^{\frac{\pi}{4}} \int_0^{\sqrt{\cos 2\theta}} \frac{r}{(1+r^2)^2} r dr d\theta \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{4}} \left[-\frac{1}{1+r^2} \right]_0^{\sqrt{\cos 2\theta}} d\theta \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{4}} 1 - \frac{1}{1+\cos 2\theta} d\theta \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{4}} 1 - \frac{1}{2} \sec^2\theta d\theta \\
 &= \frac{1}{2} \left[\theta - \frac{1}{2} \tan\theta \right]_0^{\frac{\pi}{4}} = \frac{\pi-2}{8}
 \end{aligned}$$

42.

$$\begin{aligned}
 & x, y \geq 0, x^2+y^2 \leq 1, \tan\left(\frac{x^2+y^2}{4}\right) \leq \frac{y}{x} \leq 1 \\
 & \Leftrightarrow 0 \leq r \leq 1, \tan\left(\frac{r^2}{4}\right) \leq \tan\theta \leq 1 \\
 & \Leftrightarrow 0 \leq r \leq 1, \frac{r^2}{4} \leq \theta \leq \frac{\pi}{4} \\
 & \iint_D 1 dx dy = \iint_D r dr d\theta \\
 &= \int_0^1 \int_{\frac{r^2}{4}}^{\frac{\pi}{4}} r dr d\theta \\
 &= \int_0^1 r \left(\frac{\pi}{4} - \frac{r^2}{4} \right) dr \\
 &= \frac{1}{8} \left(\pi - \frac{1}{2} \right)
 \end{aligned}$$

43.

$$\begin{aligned}
 & \iint_B \sin(16x^2+9y^2) dA \quad (x = \frac{1}{4}u, y = \frac{1}{3}v, J = \frac{1}{12}) \\
 &= \frac{1}{12} \iint_{B'} \sin(u^2+v^2) du dv \quad (R': u, v > 0, u^2+v^2 \leq 1) \\
 &= \frac{1}{12} \int_0^{\frac{\pi}{2}} \int_0^1 r \sin(r^2) dr d\theta \\
 &= \frac{1}{24} \int_0^{\frac{\pi}{2}} \int_0^1 2r \sin(r^2) dr d\theta \\
 &= \frac{1}{24} \cdot \frac{\pi}{2} \times [-\cos(r^2)]_0^1 = \frac{\pi}{48} (1 - \cos 1)
 \end{aligned}$$

44.

$$\begin{aligned}
 & x^2 - xy + y^2 \leq 2 \Leftrightarrow (x, y) \left(\frac{1}{2} - \frac{1}{2} \right) \left(\frac{x}{y} \right) \leq 2 \\
 & A = \left(\frac{1}{2} - \frac{1}{2} \right) \left(\frac{x}{y} \right) \text{는 각 행의 합이 } \frac{1}{2} \text{ 이므로 고유치 } \frac{1}{2} \text{ 가 존재하고} \\
 & tr A = 2 \text{ 이므로 나머지 고유치는 } \frac{3}{2} \text{ 이다.}
 \end{aligned}$$

또한, 각각에 대응되는 고유벡터는 $\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ 이므로

주축정리를 이용하면 $\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}$ 이다.

따라서 영역은 $D: (x, y) \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \leq 2$

$$\Rightarrow D^*: (X, Y) \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} \leq 2$$

$$\Rightarrow D^*: \frac{1}{2} X^2 + \frac{3}{2} Y^2 \leq 2 \text{ 이고}$$

$$\iint_D (x^2 - xy + y^2) dA$$

$$= \iint_D (x, y) \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} dA$$

$$= \iint_{D^*} \frac{1}{2} X^2 + \frac{3}{2} Y^2 dXdY \quad (\because |J|=1)$$

$$= \frac{2}{\sqrt{3}} \iint_{u^2+v^2 \leq 2} (u^2+v^2) du dv \quad (\because X = \sqrt{2}u, Y = \sqrt{\frac{2}{3}}v)$$

$$= \frac{2}{\sqrt{3}} \int_0^{2\pi} \int_0^{\sqrt{2}} r^2 r dr d\theta$$

$$= \frac{4\sqrt{3}\pi}{3}$$

[다른 풀이]

$$D: x^2 - xy + y^2 \leq 2 \Leftrightarrow D: \left(x - \frac{1}{2}y\right)^2 + \left(\frac{\sqrt{3}}{2}y\right)^2 \leq 2$$

$$\iint_D (x^2 - xy + y^2) dA$$

$$= \iint_D \left(x - \frac{1}{2}y\right)^2 + \left(\frac{\sqrt{3}}{2}y\right)^2 dA$$

$$= \frac{2}{\sqrt{3}} \iint_{u^2+v^2 \leq 2} (u^2+v^2) du dv \quad (\because x - \frac{1}{2}y = u, \frac{\sqrt{3}}{2}y = v, |J| = \frac{2}{\sqrt{3}})$$

$$= \frac{2}{\sqrt{3}} \int_0^{2\pi} \int_0^{\sqrt{2}} r^2 r dr d\theta$$

$$= \frac{4\sqrt{3}\pi}{3}$$

[다른 풀이]

$$D: (x, y) \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \leq 2 \text{ 일 때,}$$

$$A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \text{의 행렬식이 } \det A = \frac{3}{4} \text{ 이므로 공식에 의해}$$

$$\iint_D (x^2 - xy + y^2) dA$$

$$= \frac{1}{\frac{\sqrt{3}}{2}} \iint_{u^2+v^2 \leq 2} (u^2+v^2) du dv$$

$$= \frac{2}{\sqrt{3}} \int_0^{2\pi} \int_0^{\sqrt{2}} r^2 r dr d\theta$$

$$= \frac{4\sqrt{3}\pi}{3} \text{ 이다.}$$

45.

$$(x, y) \begin{pmatrix} 7 & 3\sqrt{3} \\ 3\sqrt{3} & 13 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \text{ 이고 } A = \begin{pmatrix} 7 & 3\sqrt{3} \\ 3\sqrt{3} & 13 \end{pmatrix} \text{의 고유치와 고유벡터를 구하면}$$

$$\begin{vmatrix} 7-\lambda & 3\sqrt{3} \\ 3\sqrt{3} & 13-\lambda \end{vmatrix} = (7-\lambda)(13-\lambda) - 27 = (\lambda-16)(\lambda-4) = 0$$

이므로 고유값은 $\lambda = 16, 4$ 이다.

$$\text{고유값에 대한 각각의 고유벡터는 } \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} -\sqrt{3} \\ 1 \end{pmatrix} \text{ 이므로}$$

$$\text{변수변환은 주축정리를 이용하면 } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} \text{ 이다.}$$

따라서

$$7x^2 + 6\sqrt{3}xy + 13y^2 = 16 \Rightarrow \frac{X^2}{4} + Y^2 = 1 \text{ 이다.}$$

$$|J| = \left| \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & 1 \end{pmatrix} \right| = \frac{1}{4} + \frac{3}{4} = 1 \text{ 이므로}$$

$$\iint_R xy dA$$

$$= \iint_{R^*} \left(\frac{1}{2}X - \frac{\sqrt{3}}{2}Y\right) \left(\frac{\sqrt{3}}{2}X + \frac{1}{2}Y\right) dXdY \quad (\text{단, } R^*: X^2 + \frac{Y^2}{4} \leq 1)$$

$$= \frac{1}{4} \iint_{R^*} \sqrt{3}X^2 - 2XY - \sqrt{3}Y^2 dXdY \quad (X=u, Y=2v)$$

$$= \frac{1}{4} \iint_{R^{**}} (\sqrt{3}u^2 - 4uv - 4\sqrt{3}v^2) 2du dv \quad (\text{단, } R^{**}: u^2 + v^2 \leq 1)$$

$$= \frac{1}{2} \int_0^{2\pi} \int_0^1 (\sqrt{3}r^2 \cos^2\theta - 4r^2 \cos\theta \sin\theta - 4\sqrt{3}r^2 \sin^2\theta) dr d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} \left(\sqrt{3} \frac{1}{4} \cos^2\theta - \cos\theta \sin\theta - \sqrt{3} \sin^2\theta \right) d\theta$$

$$= \frac{1}{2} \left(\frac{\sqrt{3}}{4} \times \frac{1}{2} \times \frac{\pi}{2} - \sqrt{3} \times \frac{1}{2} \times \frac{\pi}{2} \right) \times 4$$

$$= \pi \left(-\frac{3\sqrt{3}}{8} \right) = -\frac{3\sqrt{3}}{8} \pi$$

[다른 풀이]

$$7x^2 + 6\sqrt{3}xy + 13y^2 = 16$$

$$\Leftrightarrow \left(x + \frac{3\sqrt{3}}{7}y\right)^2 + \left(\frac{8}{7}y\right)^2 \leq \frac{16}{7}$$

$$\iint_R xy dA$$

$$= \iint_{u^2+v^2 \leq \frac{16}{7}} \left(u - \frac{3\sqrt{3}}{8}v\right) \left(\frac{7}{8}v\right) du dv$$

$$\left(\because x + \frac{3\sqrt{3}}{7}y = u, \frac{8}{7}y = v, |J| = \frac{7}{8}\right)$$

$$= \frac{49}{64} \int_0^{2\pi} \int_0^{\frac{4}{\sqrt{7}}} \left(\cos\theta - \frac{3\sqrt{3}}{8}\sin\theta\right) (\sin\theta) dr d\theta$$

$$= \frac{49}{64} \int_0^{2\pi} \frac{1}{4} \left[\frac{r^4}{4}\right]_0^{\frac{4}{\sqrt{7}}} (\cos\theta - \frac{3\sqrt{3}}{8}\sin\theta) (\sin\theta) d\theta$$

$$= \int_0^{2\pi} \left(\cos\theta - \frac{3\sqrt{3}}{8}\sin\theta\right) (\sin\theta) d\theta$$

$$= \int_0^{2\pi} \sin\theta \cos\theta - \frac{3\sqrt{3}}{8} \sin^2\theta d\theta$$

$$= \frac{1}{2} [\sin^2\theta]_0^{2\pi} - \frac{3\sqrt{3}}{8} \frac{1}{2} \pi \times 4 \quad (\text{와르리스 정리})$$

$$= -\frac{3\sqrt{3}}{8} \pi$$

46.

두 평면 $x = \sqrt{3}y$, $x = y$ 그리고 타원체 $\frac{x^2}{16} + \frac{y^2}{16} + \frac{z^2}{4} = 1$ 로 둘러싸인 제 1 팔분 공간 상의 입체의 부피를 V 라고 할 때, 다음과 같다.

$$V = \iint_D \sqrt{4 - \frac{x^2}{4} - \frac{y^2}{4}} dA \quad (\text{단, } D: x^2 + y^2 \leq 16, \frac{1}{\sqrt{3}}x \leq y \leq x)$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \int_0^4 \sqrt{4 - \frac{1}{4}r^2} r dr d\theta \text{ 이다.}$$

$$\text{따라서 } \alpha = \frac{\pi}{6}, \beta = \frac{\pi}{4}, F(r) = r\sqrt{4 - \frac{1}{4}r^2} \text{ 이므로}$$

$$(\beta - \alpha)F\left(\frac{b}{2}\right) = \left(\frac{\pi}{4} - \frac{\pi}{6}\right)F(2) = \frac{1}{12}\pi \cdot 2\sqrt{4-1} = \frac{\sqrt{3}}{6}\pi$$

47.

$$\begin{aligned}
& \int_0^{\pi} \int_0^{\infty} (x+2y)^2 f(x) f(y) dx dy \\
&= \int_0^{\pi} \int_0^{\infty} (x+2y)^2 \cdot \frac{1}{2\pi} e^{-\frac{x^2+y^2}{2}} dx dy \\
&= \frac{1}{2\pi} \int_0^{\frac{\pi}{2}} \int_0^{\infty} (r \cos \theta + 2r \sin \theta)^2 \cdot e^{-\frac{r^2}{2}} \cdot r dr d\theta \\
&= \frac{1}{2\pi} \int_0^{\frac{\pi}{2}} (\cos^2 \theta + 4 \cos \theta \sin \theta + 4 \sin^2 \theta) d\theta \cdot \int_0^{\infty} r^3 e^{-\frac{r^2}{2}} dr \\
&= \frac{1}{2\pi} \left(\frac{1}{2} \cdot \frac{\pi}{2} + [-\cos 2\theta]_0^{\frac{\pi}{2}} + 4 \cdot \frac{1}{2} \cdot \frac{\pi}{2} \right) \\
&\quad \times \int_0^{\infty} \frac{1}{2} t e^{-\frac{1}{2} t^2} dt \quad (\because r^2 = t, 2r dr = dt) \\
&= \frac{1}{2\pi} \left(\frac{\pi}{4} + \pi \right) \times \left[-t e^{-\frac{t}{2}} - 2 e^{-\frac{t}{2}} \right]_0^{\infty} \\
&= \frac{1}{2\pi} \left(\frac{5}{4} \pi + 2 \right) \times 2 = \frac{5}{4} + \frac{2}{\pi}
\end{aligned}$$

48.

$u = x^2 - y^2$, $v = 2xy$ 에 대해

$|J| = \left| \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix} \right| = 4(x^2 + y^2)$ 이다.

$$\begin{aligned}
& \iint_{G(A)} \frac{1}{\sqrt{u^2 + v^2}} du dv \\
&= \iint_A \frac{1}{\sqrt{(x^2 - y^2)^2 + (2xy)^2}} \cdot 4(x^2 + y^2) dx dy \\
&= \iint_A 4 dx dy \\
&= 4 \int_0^{\frac{\pi}{2}} \int_{\frac{1}{2}}^1 r dr d\theta \\
&= 4 \cdot \frac{1}{2} \cdot \frac{\pi}{2} \cdot \frac{3}{4} = \frac{3}{4} \pi
\end{aligned}$$

49.

P_1 의 내부이면서 P_2 의 외부인 영역 중 $D: 1 < x < 2, 0 < y < x$ 라 하면

$\iint_R \frac{1}{(x^2 + y^2)^2} dA = 8 \iint_D \frac{1}{(x^2 + y^2)^2} dA$ 가 성립한다.

따라서

$$\begin{aligned}
& \iint_R \frac{1}{(x^2 + y^2)^2} dA \\
&= 8 \int_0^{\frac{\pi}{4}} \int_{\sec \theta}^{2 \sec \theta} \frac{1}{r^3} dr d\theta \\
&= 8 \int_0^{\frac{\pi}{4}} \left[-\frac{1}{2} \left(\frac{1}{r^2} \right) \right]_{\sec \theta}^{2 \sec \theta} d\theta \\
&= -4 \int_0^{\frac{\pi}{4}} \frac{1}{4} \cos^2 \theta - \cos^2 \theta d\theta \\
&= -4 \int_0^{\frac{\pi}{4}} \frac{3}{4} \cos^2 \theta d\theta \\
&= -3 \int_0^{\frac{\pi}{4}} \cos^2 \theta d\theta \\
&= 3 \left[\frac{\theta}{2} + \frac{1}{4} \sin 2\theta \right]_0^{\frac{\pi}{4}} \\
&= \frac{3}{8} \pi + \frac{3}{4} \text{ 이다.}
\end{aligned}$$

50.

$$\begin{aligned}
& \int_{-\infty}^{\infty} e^{-2x^2 + 8x} dx \\
&= \int_{-\infty}^{\infty} e^{-2(x-2)^2 + 8} dx \\
&= e^8 \int_{-\infty}^{\infty} e^{-2(x-2)^2} dx \quad (\because \sqrt{2}(x-2) = t) \\
&= \frac{e^8}{\sqrt{2}} \int_{-\infty}^{\infty} e^{-t^2} dt = e^8 \frac{\sqrt{\pi}}{\sqrt{2}}
\end{aligned}$$